On perfect factoriangular numbers

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with an appendix by Sawian Jaidee

Abstract

A factoriangular number is a number of the form $Ft_n = T_n + n!$, where $T_n = n(n+1)/2$ is the nth triangular number. We show that the set of $n \le x$ such that Ft_n is perfect has counting function at most $x^{o(1)}$ as $x \to \infty$.

1 Introduction

For positive integers n, let $Ft_n = T_n + n!$, where $T_n = n(n+1)/2$, be the nth factoriangular (the sum between the nth triangular number and the factorial of n). These numbers were first studied in [2], where several conjectures concerning them appear. For example, it was asked if $2 = Ft_1$, $Ft_2 = 5$ and $Ft_4 = 34$ are the only factoriangular numbers which are also Fibonacci numbers. This was confirmed in [3]. A different conjecture from [2] concerning factoriangular numbers which are members of the Pell sequence was confirmed in [4]. In [2], it was also asked to prove that Ft_n is never perfect; that is, equal to the sum of its proper divisors. While we cannot prove that, we prove the following theorem.

Theorem 1. *The number of $n \le x$ such that Ft_n is perfect is at most $x^{o(1)}$.*

Our result is similar to Wirsing's theorem [6] concerning the count of perfect numbers $n \le x$ which states that this count is at most $x^{o(1)}$ for $x \to \infty$. In Wirsing's result, the quantity $o(1)$ in the exponent stands for a function of order $O(1/\log\log x)$. As the reader will see, in our theorem the $o(1)$ can be taken to have this same order.

2 The proof

If $n \ge 6$ and $n+1$ is not prime, then $n+1 \mid n!$. Indeed, writing $n+1 = ab$ with $a \le b$, we have that if $a < b$ then a and $b \le (n+1)/2 < n$ are distinct positive integers less than n so $n+1 = ab \mid n!$. If $a = b = \sqrt{n+1}$, then a, $2a < n$ are distinct positive integers less than n, so again $n+1 = a^2 \mid n!$. Thus, T_n divides $n!$, so also Ft_n, except when $n+1$ is prime.

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Lemma 1. *Ft_n is not perfect if $n+1$ is prime.*

Proof. One checks by hand that this is so for $n = 1, 2$. Let $p = n + 1 \geq 5$ be prime. Then $n = p - 1$ and

$$Ft_n = \frac{(p-1)}{2}(p + 2(p-2)!).$$

Note that every prime factor dividing $p+2(p-2)!$ must be larger than p (indeed, p does not divide $p + 2(p-2)!$ and if $q < p$, then $q \leq p - 2$ so q divides $2(p-2)!$ but not p). If Ft_n is perfect then $\sigma((p-1)/2) < p-1$, since $(p-1)/2$ is a proper divisor of Ft_n. But $\sigma((p-1)/2)$ divides

$$\sigma\left(\frac{p-1}{2}\right)\sigma(p + 2(p-2)!) = \sigma(Ft_n) = 2Ft_n = (p-1)(p + 2(p-2)!)$$

and being smaller than $p - 1$ is coprime to the factor $p + 2(p-2)!$. The only possibility is $\sigma((p-1)/2) = (p-1)/2$, which is false. $\square$

From now on, we assume that $n \geq 6$ and $n + 1$ is not prime, so $n + 1 \mid n!$. Since $n + 1$ is coprime to n, so that $n + 1$ divides $(n-1)!$, M_n appearing below is an (odd) integer. We write

$$Ft_n = T_n\left(1 + \frac{2(n-1)!}{n+1}\right) = \frac{n(n+1)}{2}M_n.$$

We take a look at the prime factors of M_n.

Lemma 2. *Assume n is sufficiently large. All prime factors of M_n are larger than n except when $(n+1)/2 = q$ is a prime in which case $q \mid M_n$ but all other prime factors of M_n are larger than n.*

Proof. We first show that if $q \leq n - 1$ is prime and $q \neq \frac{n+1}{2}$, then

$$q \mid \frac{2(n-1)!}{n+1}, \tag{0}$$

so that $q \nmid M_n$. The relation (0) is immediate if n is odd: In that case, $q\frac{n+1}{2} \mid (n-1)!$, which is equivalent to (0). Suppose now that n is even. We may assume that $q \mid n + 1$, since (0) is obvious otherwise. Then $q \leq (n+1)/3$, so that $q^2 \mid q \cdot 2q \mid (n-1)!$. So in order for (0) to fail, we must have $q^k \parallel n + 1$ for an integer $k \geq 2$. Then $q \leq \sqrt{n+1}$ and $k \leq \log(n+1)/\log 3 < n^{1/3}$ (recalling that n is large for all of this). But then q, $2q$, $\ldots$, $(k+1)q$ are all at most $n - 1$, so that $q^{k+1} \mid (n-1)!$, implying that (0) holds.

If n is odd and $(n+1)/2 = q$ is prime, then $n = 2q - 1$ and

$$M_n = 1 + (q-1)!(q+1)\cdots(2q-2) \equiv 1 + (q-1)!(q-2)! \pmod q \equiv 0 \pmod q$$

by Wilson's theorem. Finally, Wilson's theorem again implies that if $n = p$, then

$$M_p = 1 + \frac{2(p-1)!}{p+1} \equiv 1 - 2 \pmod p \equiv -1 \pmod p$$

is not a multiple of p. Hence, every prime factor of M_n is larger than n except when $(n+1)/2 = q$ is prime case in which also $q \mid M_n$. $\square$

Lemma 3. *We have*

$$\frac{\sigma(M_n)}{M_n} = 1 + O\left(\frac{\log\log n}{\log n}\right).$$

Proof. We have $M_n < n^n$ for $n \geq 3$. Let $\omega(m)$ be the number of distinct prime factors of m. Since $\omega(m) \leq (1+o(1))\log m/\log\log m$ for $m \to \infty$, we have that $\omega(M_n) < 2n$ holds for all large enough values of n. Let $p_1 < p_2 < \cdots < p_k$ be all the prime factors of M_n. Write

$$M_n = \prod_{i=1}^{k} p_i^{a_i}.$$

We have

$$\frac{\sigma(M_n)}{M_n} = \prod_{i=1}^{k}\left(1 + \frac{1}{p_i} + \cdots + \frac{1}{p_i^{a_i}}\right) = \exp\left(\sum_{i=1}^{k}\left(\frac{1}{p_i} + \mathcal{O}\left(\frac{1}{p_i^2}\right)\right)\right). \tag{1}$$

By Lemma 2, we have either $p_1 > n$ or $p_1 = (n+1)/2$ and $p_2 > n$. We also have that $\omega(M_n) < 2n$ holds for large n. By the prime number theorem, the interval $(n, 3n\log n)$ contains $\pi(3n\log n) - \pi(n) > 2n$ primes for large n. This shows that for large n we have

$$\begin{aligned}
\sum_{i=1}^{k}\left(\frac{1}{p_i} + O\left(\frac{1}{p_i^2}\right)\right) &\leq \frac{1}{(n+1)/2} + \sum_{n<p<3n\log n}\frac{1}{p} + O\left(\frac{1}{n}\right)\\
&\leq \log\log(3n\log n) - \log\log n + O\left(\frac{1}{\log n}\right)\\
&= \log(\log n + \log(3\log n)) - \log\log n + O\left(\frac{1}{\log n}\right)\\
&\leq O\left(\frac{\log\log n}{\log n}\right).
\end{aligned} \tag{2}$$

In the above, we used the Mertens estimate

$$\sum_{p\leq t}\frac{1}{p} = \log\log t + \kappa + O\left(\frac{1}{\log t}\right)$$

valid for all $t \geq 2$ with some constant κ. The result follows by inserting (2) into (1). $\square$

We are now ready to address the case when $(n+1)/2 = q$ is prime.

Lemma 4. *For all large enough n, it is impossible for $(n+1)/2 = q$ to be prime and Ft_n to be perfect.*

Proof. Suppose n is large, that $\frac{n+1}{2} = q$ is prime, and that Ft_n is perfect. Then

$$Ft_n = n(qM_n),$$

where every prime factor of qM_n is either q or larger than n. Note that n is coprime to qM_n. Since Ft_n is perfect and $n \mid Ft_n$, we get that $\sigma(n) < 2n$ is a divisor of $2nqM_n$. Assume first that $\sigma(n)$ is prime. Then $n = p^\alpha$ for some odd prime p. Thus,

$$\frac{\sigma(n)}{n} \leq \frac{n}{\phi(n)} = \frac{p}{p-1} \leq \frac{3}{2},$$

so, by Lemma 3, we get

$$2 = \frac{\sigma(Ft_n)}{Ft_n} = \frac{\sigma(n)}{n}\frac{\sigma(qM_n)}{qM_n} \leq \frac{\sigma(n)}{n}\frac{\sigma(q)}{q}\frac{\sigma(M_n)}{M_n} \leq \frac{3}{2}\left(1 + O\left(\frac{\log\log n}{\log n}\right)\right), \tag{3}$$

which is false for large n. Assume next that $\sigma(n)$ is not prime. It then follows that all prime factors of $\sigma(n)$ are at most n. Hence, $\sigma(n) \mid 2nq^s$, where q^s is the highest power of q in qM_n. Hence, $\sigma(n) = q^t\lambda$ where $0 \leq t \leq s$ and λ is a divisor of $2n$. The case $t = 0$ is impossible since it leads to $\sigma(n)$ being a divisor of $2n$ which is in the interval $(n, 2n)$, which does not exist. The case $t > 1$ is also impossible, as then $\sigma(n) \geq q^2 > 2n$. Finally suppose $t = 1$. Then $2n > \sigma(n) = \lambda\frac{n+1}{2} > n$, forcing $\lambda \in \{2, 3\}$. Thus, $\sigma(n)/n \leq (3/2) + O(1/n)$, and we get again a contradiction as in (3). □

From now on, we may assume that $(n+1)/2$ is not prime. Thus

$$Ft_n = T_nM_n,$$

where all prime factors of M_n exceed n and (recalling that $n+1$ is not prime) all prime factors of T_n are at most n.

Lemma 5. *Ft_n cannot be an even perfect number.*

Proof. One checks directly that no Ft_n is perfect when $n < 6$, so we may assume that $n \geq 6$. If $n+1$ is prime, then Ft_n is not perfect (Lemma 1). Otherwise, as explained after Lemma 1, our Ft_n is of the form

$$T_nM_n,$$

where M_n is odd. But even perfect numbers are of the form $2^{p-1}P$, where $P = 2^p - 1$ is prime. If Ft_n is even perfect, then $T_n = 2^{p-1}$ is a power of 2, which is impossible for $n \geq 2$. □

Remark 1. *The fact that a factoriangular number is not even perfect was also noted by Ioulia Baoulina in [1].*

From now on, we assume that Ft_n is an odd perfect number. By a result of Euler, it must be of the form qw^2, where q is a prime. Thus,

$$Ft_n = T_nM_n = qw^2.$$

It follows that either $q \mid T_n$ or $q \mid M_n$.

Lemma 6. *The number of positive integers $n \leq x$ such that Ft_n is odd perfect of the form qw^2 where q is a prime dividing M_n is $O(\log x)$.*

Proof. Since T_n and M_n are relatively prime, in this case $T_n = v^2$, for some divisor v of w. We get

$$(2n+1)^2 - 2(2v)^2 = 1.$$

Thus, $2n+1 = P_m$ for some index m, where $(P_m, Q_m) = (P, Q)$ is the mth solution of the Pell equation $P^2 - 2Q^2 = \pm 1$. We have $(P_1, Q_1) = (1, 1)$ and

$$P_m + \sqrt{2} Q_m = (1+\sqrt{2})^m.$$

Since $n \le x$, it follows that $P_m \le 2x+1$, which leads to $m = O(\log x)$. $\square$

From now on, we assume that $q \mid T_n$, so $T_n = qu^2$ for some divisor u of w. We write

$$T_n = n\left(\frac{n+1}{2}\right), \quad (n+1)\frac{n}{2}$$

according to whether n is odd or even, respectively. If n is even, then $\sigma(n/2) < n$ and if n is odd then $\sigma((n+1)/2) < n+1$, so $\sigma((n+1)/2) \le n$. So, $\sigma(n/2)$ for even n and $\sigma((n+1)/2)$ for odd n have all prime factors at most n. If $\sigma(n+1)$ (for n even) or $\sigma(n)$ (for n odd) is not a prime number, then all prime factors of $\sigma(n+1)$ (for n even) and $\sigma(n)$ (for n odd) are also at most n. In either of these cases, $\sigma(n(n+1)/2)$ has all prime factors at most n. Since $\sigma(n(n+1)/2)$ divides $\sigma(T_n M_n) = n(n+1)M_n$, we get that $\sigma(n(n+1)/2)$ divides $n(n+1)$. But $n(n+1)/2 < \sigma(n(n+1)/2) < n(n+1)$, so this is impossible. Hence, it must be that when n is even we have $\sigma(n+1)$ is a prime and when n is odd we have that $\sigma(n)$ is a prime. Thus, either $n+1$ or n equal p^α for some odd prime p such that $\sigma(p^\alpha) = (p^{\alpha+1}-1)/(p-1)$ is prime. Further, α must be even otherwise $(p^{\alpha+1}-1)/(p-1)$ is a multiple of $p+1$ so it cannot be prime. Hence, we have either

$$n+1 = p^\alpha, \quad n/2 = qu_1^2, \quad \text{when } n \equiv 0 \pmod 2,$$

or

$$n = p^\alpha, \quad (n+1)/2 = qu_1^2, \quad \text{when } n \equiv 1 \pmod 2,$$

where in both cases $u = u_1 p^{\alpha/2}$. The case n even leads to

$$qu_1^2 = \frac{n}{2} = \frac{p^\alpha - 1}{2},$$

which is not possible since α is even, so $p^\alpha \equiv 1 \pmod 4$, therefore the right-hand side above is even. Thus, $n = p^\alpha$ where p is an odd prime.

We need one more lemma.

Lemma 7. *Let q_1 be the smallest prime factor of $\frac{n(n+1)}{2}$. Then $q_1 \le (\log n)^{2/3}$ for large n.*

Proof. Let $\ell = \omega(n(n+1)/2)$ and write $q_1 < q_2 < \cdots < q_\ell$ for all the prime factors of $n(n+1)/2$. Note that $\ell \ge 2$ since p and q are among these primes. Write

$$\frac{n(n+1)}{2} = \prod_{j=1}^{\ell} q_j^{b_j}.$$

Assume $q_1 > (\log n)^{2/3}$. Then

$$\begin{aligned} 2 = \frac{\sigma(Ft_n)}{Ft_n} &= \frac{\sigma(T_n)}{T_n}\frac{\sigma(M_n)}{M_n} \\ &= \prod_{j=1}^{\ell}\left(1 + \frac{1}{q_j} + \cdots + \frac{1}{q_j^{b_j}}\right)\left(1 + O\left(\frac{\log\log n}{\log n}\right)\right) \\ &= \exp\left(\sum_{j=1}^{\ell}\left(\frac{1}{q_j} + O\left(\frac{1}{q_j^2}\right)\right)\right)\left(1 + O\left(\frac{\log\log n}{\log n}\right)\right). \end{aligned} \tag{4}$$

Since $T_n < 2n^2$, it follows that $\ell \le 3\log n/\log\log n$ for large n. By the prime number theorem, the interval $((\log n)^{2/3}, 4\log n)$ contains

$$\pi(4\log n) - \pi((\log n)^{2/3}) > \frac{3.5\log n}{\log\log n} - (\log n)^{2/3} > \frac{3\log n}{\log\log n} > \ell$$

primes for n sufficiently large. This shows that

$$\begin{aligned} \sum_{j=1}^{\ell}\frac{1}{q_j} &\le \sum_{(\log n)^{2/3} < q < 4\log n}\frac{1}{q} \\ &= \log\log(4\log n) - \log\log((\log n)^{2/3}) + O\left(\frac{1}{\log\log n}\right) \\ &= \log(3/2) + O\left(\frac{1}{\log\log n}\right). \end{aligned} \tag{5}$$

Inserting (5) into (4), we get

$$2 \le \exp\left(\log(3/2) + O\left(\frac{1}{\log\log n}\right)\right)\left(1 + O\left(\frac{\log\log n}{\log n}\right)\right) = \frac{3}{2} + O\left(\frac{1}{\log\log n}\right),$$

which is false for large n. □

We now put everything together using arguments of Pollack [5] and Wirsing [6].

Theorem 2. *The number of $n \le x$ such that the factoriangular Ft_n is perfect is at most $x^{O(1/\log\log x)}$.*

Proof. From the previous arguments, we get that aside from the family of positive integers n arising as $2n+1 = P_m$ for some m, where (P_m, Q_m) is the mth solution to the Pell equation $P^2 - 2Q^2 = \pm 1$, which has a counting function of size $O(\log x)$, all the other potential n's have $n = p^\alpha$ for some odd prime p and even number α. Note that if p is known, then $\alpha = O(\log x)$. Thus, once we know p, we just multiply the number of choices for p by a factor of $\log x$ and get a bound on the number of such possible n's.

From the last paragraph, we may assume that $p > (\log x)^{2/3}$. Now let B be the largest divisor of $n(n+1)/2$ composed of primes not exceeding $(\log x)^{2/3}$. Then B divides $(n+1)/2$ and Lemma 7 gives us that $B > 1$. Since the exponent on each prime dividing B is at most

$\log x/\log 2$ while $\pi((\log x)^{2/3}) \le (\log x)^{2/3}$, the number of possibilities for B is bounded above by

$$\left(1+\frac{\log x}{\log 2}\right)^{(\log x)^{2/3}} \le x^{O(1/\log\log x)}.$$

We put $B_1 = B$ and proceed to define a sequence $B_1, B_2, B_3, \ldots$ inductively. Assume $B_1, \ldots, B_s$ have already been selected, where

$$B_1 \mid \cdots \mid B_s \mid \frac{n+1}{2}, \quad \text{each} \quad \omega(B_i) \ge i, \quad \text{and each} \quad \gcd\left(B_i, \frac{(n+1)/2}{B_i}\right) = 1. \tag{6}$$

Since $B_s > 1$ is a proper divisor of the perfect number Ft_n, we have that $B_s < \sigma(B_s) < 2B_s$. Hence, $\sigma(B_s) \nmid 2B_s$, and there is a prime appearing in $\sigma(B_s)$ to a higher power than in $2B_s$. Let q_{s+1} be the least such prime. If $q_{s+1} = p$, we terminate the B-sequence with $B_1, \ldots, B_s$.

If $q_{s+1} \neq p$, we continue as follows. We know that $\sigma(B_s)$ divides $\sigma(Ft_n) = n(n+1)M_n$. But $\sigma(B_s) < 2B_s \le n+1$, so that every prime factor of $\sigma(B_s)$ is smaller than every prime factor of M_n. Hence,

$$\sigma(B_s) \quad \text{divides} \quad n(n+1) = 2B_s \cdot p^{\alpha} \cdot \frac{(n+1)/2}{B_s}.$$

Since q_{s+1} appears to a higher power in $\sigma(B_s)$ than in $2B_s$, and $q_{s+1} \neq p$, we must have that q_{s+1} divides $\frac{(n+1)/2}{B_s}$. Then $q_{s+1} \nmid B_s$ and $q_{s+1} \mid (n+1)/2$. Let $q_{s+1}^{\beta_{s+1}}$ be the highest power of q_{s+1} occurring in $(n+1)/2$ and set $B_{s+1} := B_s q_{s+1}^{\beta_{s+1}}$. Then (6) holds with s replaced by $s+1$.

Let $B_1, B_2, \ldots, B_t$ be the full list of terms of the B-sequence. Necessarily, $t \le \omega((n+1)/2)$. According to the condition of termination, $p \mid \sigma(B_t)$. Since $\sigma(B_t) \le x$, given B_t there are $O(\log x)$ choices for p and (by the comments at the start of the proof) $O((\log x)^2)$ choices for n. To count the number of possibilities for

$$B_t = B q_2^{\beta_2} \cdots q_t^{\beta_t},$$

observe that B_t is determined by B and the exponent sequence $\beta_2, \beta_3, \ldots, \beta_t$. (The point is that B_s determines q_{s+1}, and q_{s+1} along with β_{s+1} determines B_{s+1}. This maneuver is central in both [5] and [6].) We saw earlier that there are $x^{O(1/\log\log x)}$ choices for B. Since

$$((\log x)^{2/3})^{\beta_2+\cdots+\beta_t} \le q_2^{\beta_2} \cdots q_t^{\beta_t} \le x,$$

the exponent sequence $\beta_2, \ldots, \beta_t$ has sum

$$\beta_2 + \cdots + \beta_t \le \left\lfloor \frac{3}{2} \frac{\log x}{\log\log x} \right\rfloor =: S.$$

But the number of (possibly empty) sequences of positive integers with sum at most S is $2^S = x^{O(1/\log\log x)}$. Thus, there are $x^{O(1/\log\log x)}$ possibilities for B_t and (after absorbing bounded powers of $\log x$) also for n. This finishes the proof. $\square$

Remark 2. *This paper was prepared for publication from Florian Luca's original manuscript. P.P. checked the mathematical details, made corrections, and rewrote various components. ChatGPT 5.6 was used to audit the arguments, but all the mathematics in this paper was human-generated.*

Acknowledgements

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Appendix by Sawian Jaidee: Report on the original manuscript

2.1 General Assessment

The manuscript studies factoriangular numbers

$$Ft_n = T_n + n!,$$

where $T_n = n(n+1)/2$, and seeks to prove that the number of indices $n \le x$ for which Ft_n is a perfect number is at most $x^{O(\log\log\log x/\log\log x)}$. The topic is interesting and fits naturally into the literature on perfect numbers, factorial-type sequences, and Diophantine properties of factoriangular numbers. The proposed theorem would be a meaningful partial result toward the conjecture that no factoriangular number is perfect.

However, the current proof contains several substantial gaps, incorrect statements, sign errors, and unjustified divisibility arguments. Some of these appear repairable, but others occur at central points of the proof, particularly Lemma 4 and the final Pollack-type counting argument. Consequently, the main theorem is not established in the manuscript's present form.

2.2 Detailed Comments

- **Lemma 1:** Contains an incorrect divisibility conclusion. Even if it were correct, it may be repairable. There is also a notation error where $\sigma(p-1)$ appears instead of $\sigma((p-1)/2)$.

- **Lemma 6:** The Pell equation has the wrong sign. If $T_n = v^2$, then $n(n+1)/2 = v^2$. Consequently,
$$(2n+1)^2 - 2(2v)^2 = 1,$$
not -1. The manuscript displays
$$(2n+1)^2 - 2(2v)^2 = -1.$$
This also reverses the parity condition on the index of the Pell solutions. All solutions are
$$Q_m + \sqrt{2}P_m = (1+\sqrt{2})^{2m+1}.$$
Check this carefully. The desired $O(\log x)$ counting conclusion should still follow after correction, but the statement and proof of the lemma are presently incorrect.

- **Lemma 2, Lemma 4, Lemma 5, Lemma 7 and Theorem 2:** Recheck the proofs carefully.

- **Language and notation:** The manuscript would benefit from a thorough revision. For example, F_n, F_{t_n}, and $F_t(n)$ are used inconsistently.

2.3 Strengths

Despite the above issues, the manuscript has several strengths:

1. The central problem is natural and appears to be previously unresolved.

2. The factorization $F_n = T_n M_n$ is useful.

3. The observation that prime factors of M_n are generally larger than n is potentially powerful.

4. The estimate $\sigma(M_n)/M_n = 1 + o(1)$ is an appropriate tool for excluding various configurations.

5. The division into even and odd perfect numbers is natural.

6. The Pell-equation reduction gives an efficient count in one branch.

7. A Pollack-style recursive divisor argument is a promising route to a subpolynomial estimate.